



OPENAI SOLUTIONS ENGINEERING

From fragmented supply-chain signals to governed decisions

A proposed OpenAI platform approach for Zeiss Supply Chain Management and IT leadership



TL;DR

OpenAI becomes the governed decision layer between ZEISS supply-chain data and operational action.

| Executive chat experience

| Enterprise integration and control layer

Problem

Context is split across SAP, logistics, suppliers and files.

Platform move

A central communication interface gives leaders **one decision surface**; the API integrates tools and controls.

Pilot

Start with one high-value workflow and validate faster triage, evidence quality and governed approvals.



Status quo and validation needs

Known operating context



- Advanced manufacturing, medical technology, quality and optics domains
- Decisions draw on SAP, supplier, logistics and document context
- Value depends on trustworthy answers, not another dashboard

Discovery hypotheses to validate

- Which disruptions consume the most time?
- Where do handoffs reduce accuracy or predictability?
- Which recommendations need evidence, review and traceability?



Providing a decision framework for existing data

SAP

- POs
- Material master
- MRP exceptions

Warehouse

- EWM stock
- Goods receipts
- Production buffers

Carriers

- DHL / FedEx
- ETA changes
- Customs events

Suppliers

- Commit dates
- Capacity
- Quality status

Files

- Scorecards
- Excel trackers
- Contracts

PAIN

- Manual effort to reconstruct one trusted view
- Answer accuracy varies by source snapshot
- Low precision for lead-time and supplier risk
- Reliability depends on scarce experts

COST

- Expedite spend and buffers absorb uncertainty
- Leadership time shifts into escalation meetings
- Late mitigation creates schedule churn
- Scattered evidence increases quality exposure



Supply Chain Hub as the enterprise decision interface

Communication interface

An executive-grade conversation surface where teams ask business questions, inspect evidence and **align on next steps**.

Enterprise operating model

Role scope, source ownership, approval paths and auditability turn the app into a **governed way of working**.

System integration layer

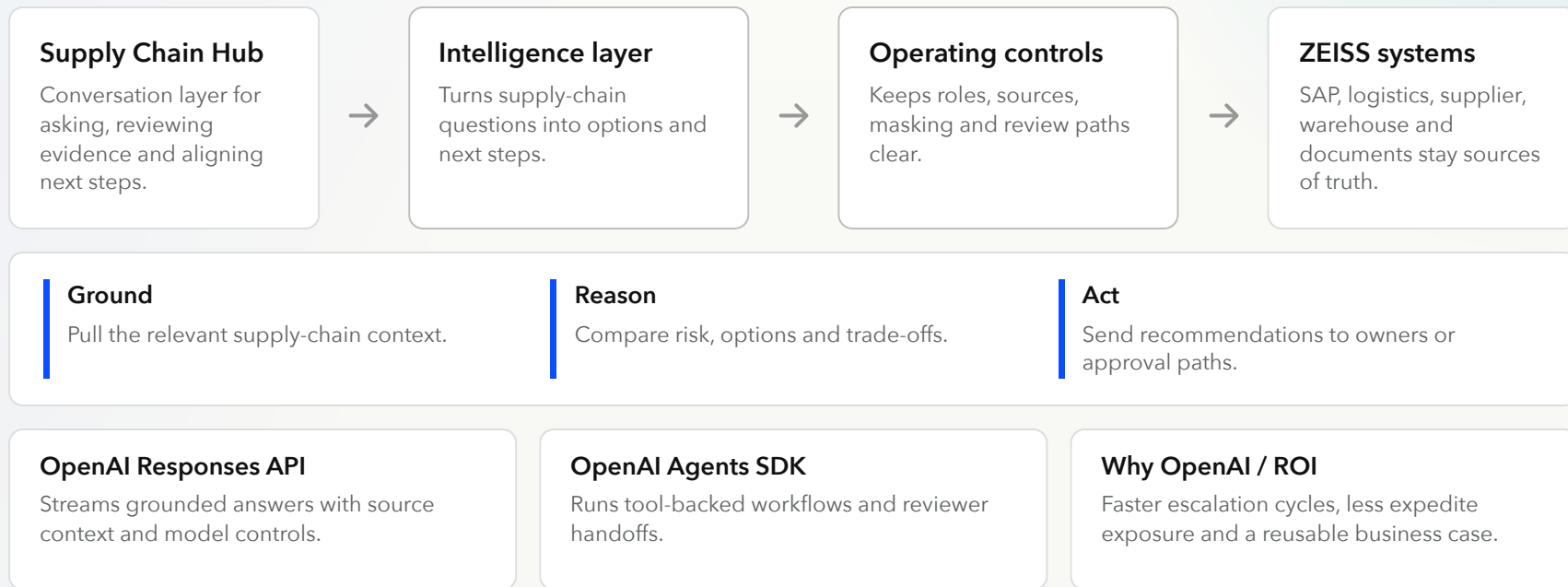
Connects ZEISS systems, documents, business tools and workflow actions, constantly **keeping these sources of truth up to date**.

BUSINESS OUTCOME

- Reduce manual effort by consolidating scattered supply-chain context into one decision interface.
- Improve decision precision with source-backed recommendations and visible assumptions.
- Give leaders a reliable path from signal to recommended action and approval.

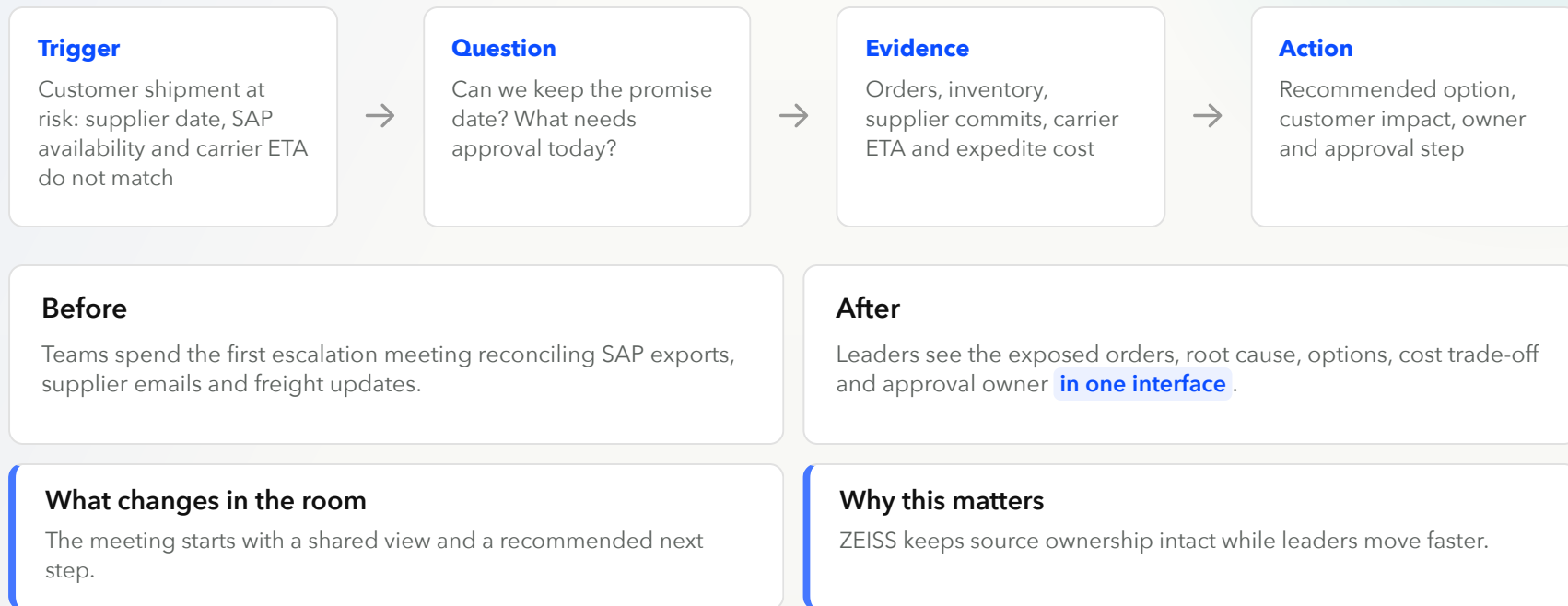


Solution overview





Executive workflow: from risk to next step



Demo



Impact validation and success criteria

The first POC must prove both business impact and operating trust: faster decisions, better evidence and controlled action paths.

1

Impact validation

Baseline

Measure current risk-review cycle time, escalation volume and manual triage hours.

Run the workflow

Use one live supply-chain risk scenario with real source boundaries and named review.

Read out value

Compare avoided effort, decision speed and quality of evidence against baseline.

2

Measurable success criteria

25%

Faster weekly risk-review cycle versus baseline.

90%

Answers cite trusted ZEISS sources and show assumptions.

80%

Target users rate outputs useful for real workflows.

100%

High-impact actions route to accountable review.



Supply Chain Hub as the business blueprint

Procurement

- Supplier negotiations
- Contract Q&A
- Spend-risk trade-offs

Quality

- Deviation triage
- Corrective action summaries
- Supplier quality signals

Manufacturing

- Line impact analysis
- Work instruction support
- Maintenance handoffs

Finance

- Working capital views
- Expedite-cost exposure
- Scenario assumptions

The repeatable blueprint is the same: conversational workspace, integration layer, enterprise context and accountable workflow ownership.



Proof of value including guardrails

Four-week proof of concept maximum, followed by a business decision on pilot scale.

WEEK 0

Align

Sponsor, workflow owner, success metrics, data boundaries.

WEEK 1

Connect

Representative sources, role scopes, approval policy.

WEEK 2

Build

App workflow, business tools, grounding and audit trail.

WEEK 3

Validate

User feedback, answer quality checks, security review.

WEEK 4

Decide

Impact readout, guardrail review, pilot recommendation.

What is needed to get started: executive sponsorship, access to representative systems and documents, a named business process owner and agreement on operating boundaries for the first controlled pilot.

Annex



Behind the scenes - actual Supply Chain Hub stack

App experience

- Next.js App Router, React and strict TypeScript
- `app/supply-chain-app.tsx` uses AI SDK `useChat`
- Persona, model, thinking and source-selection controls

OpenAI chat path

- `/api/chat` validates requests and builds server context
- `@ai-sdk/openai` calls `openai.responses( ... )`
- Vercel AI SDK streams output; deterministic demo stream remains fallback

Agentic action path

- `/api/actions` checks workflow and persona access
- `@openai/agents` uses Agent, tool, handoff and run
- Reviewer handoffs plus trace flush; fallback results preserve demo reliability

Repo controls and runtime

- AGENTS.md: shared logic in `lib/`; trust in `auth / permissions`
- `chat-extensions.ts` is the MCP / RAG extension point
- OpenNext + Wrangler deploy to Cloudflare Workers; Vitest covers routes and UI